Bellman Conservation Law-based Large-Scale Pedestrian Flow Control via Passage Direction and Entrance-Rate Optimization in Metropolitan tourism hotspots

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Abstract—**Open scenic areas and major event venues often face systematic stampede risks caused by highly concentrated pedestrian flows, while existing large-scale crowd control methods largely rely on empirical rules and post-event reviews, making active prevention and quantitative evaluation difficult. In such scenarios, the principal operational controls are to set passage directions, including one-way, two-way, and closed states, and to regulate time-varying entrance release intensities; however, the coupled effects of these two types of controls on global pedestrian flow still lack quantitative modeling and optimization tools. To address this problem, this paper proposes a macroscopic crowd dynamics modeling and control-measure optimization method for fine-grained large-scale crowd management. The model is built upon the Hughes continuum framework. To represent operational rules in a continuum setting, local admissible direction sets are introduced to characterize discrete control states such as one-way movement, two-way movement, and channel closure, while anisotropic mobility tensors describe the geometric guidance exerted by channel layouts on pedestrian flow. On this basis, internal channel entrance flow-rate control is further introduced, enabling a control strategy to jointly specify travel directions and time-varying release intensities. To solve the resulting mixed-variable optimization problem composed of discrete direction configurations and continuous release intensities, this paper designs a Hierarchical Constrained Mixed-variable Black-box Optimization (HCMBO) algorithm for searching optimal crowd control strategies. Experiments on a simplified scenic-platform scenario show that the proposed macroscopic model can capture the complex interactions among multi-stage behavior, channel geometric guidance, and direction constraints. Under a weighted composite of efficiency, safety, load-balance, waiting, and smoothness terms, repeated experiments show that HCMBO achieves the lowest mean objective value. It reduces the objective by approximately 3.9% compared with the second-best TPE-style mixed Bayesian optimization method, and also outperforms random search, pure simulated annealing, differential evolution, and other baseline methods. The proposed method provides a quantitative and interpretable decision-support tool for crowd routing and entrance flow control in open public spaces.**

Index Terms—**Crowd dynamics modeling, crowd flow control, mixed-variable black-box optimization.**

I. Introduction

Large-scale crowd gathering poses risks not only to individual safety but also to public order, as localized congestion can escalate into stampede incidents. Effective crowd management is therefore a critical component of urban public-safety governance [1], [2]. High-density pedestrian activities, such as holiday tourism, public events, and major sports gatherings, are becoming increasingly frequent. As a result, core urban areas must accommodate pedestrian demands far exceeding daily levels within short time windows, which poses major challenges for on-site organization, risk early warning, and emergency dispatch [3], [4], [5]. Against this backdrop, scientific modeling, numerical simulation, and optimization of crowd-control strategies have become critical research problems for social safety and urban governance [6], [7]. Such methods aim to improve the efficiency of large-scale event organization while reducing local congestion risks.

Crowds in large-scale events constitute typical complex systems. Numerous individuals with heterogeneous desired speeds, destinations, and route preferences interact through multi-scale, nonlinear mechanisms in confined spaces, generating spatiotemporal patterns at the collective level [4], [8]. Macroscopically, these patterns appear as destination-oriented flow, passage convergence, bottleneck queuing, and directional responses to control regulations [9], [10]. Microscopically, individuals are influenced by surrounding density, visible paths, obstacles, and guidance facilities [11]. For open scenic areas, crowd behavior additionally exhibits pronounced phase dependence: visitors may first move toward a sightseeing platform, then tour along the main viewing direction, and subsequently choose different exit passages based on preferences [12]. These characteristics of multi-stage, multi-route coexistence with shared congestion mean that traditional static capacity analysis or single shortest-path models cannot adequately support refined management decisions.

Among various urban crowd gathering scenarios, open scenic areas, waterfront sightseeing platforms, and historic commercial districts are highly representative. Unlike enclosed venues such as stadiums and subway stations with well-defined boundaries [13], [14], such areas consist of open spaces, main tourist corridors, multiple lateral connecting passages, and return flow lines. In these settings, visitors typically undergo a continuous behavioral chain of "entry–touring along the main corridor–choosing passages to leave–returning" [15]. Their movement is therefore governed jointly by spatial geometry, control regulations, and historical behavioral preferences [16]. Taking the Shanghai Nanjing East Road–Bund area as a concrete example, visitors approach the Bund from Nanjing East Road, ascend to the Bund sightseeing platform via multiple stepped passages, tour along the waterfront, then exit through several southern passages and return along lower streets. This scenario can be abstracted as the typical "Bund sightseeing platform–multi-stepped passage" crowd flow scenario. In such

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(a) (b)

Fig. 1. Field-control context of the Shanghai Nanjing East Road–Bund area. (a) Large-scale holiday crowd under one-way passage organization; image source: [17]. (b) Entrance-capacity control at a main visitor inflow location; image source: [18].

scenarios, management difficulties appear in three aspects. First, multiple passages simultaneously serve entry, exit, and diversion functions, so local direction settings can alter the overall flow organization. Second, behavioral transitions between the main tourist corridor and lateral passages make single origin–destination models insufficient. Third, local congestion and global route choice form feedback loops, where control changes in one passage may redistribute loads in adjacent areas.

Existing large-scale event security management practices typically integrate system dynamics, system coupling, and system safety theory, combined with high-risk zone delineation and historical crowd behavior patterns, to formulate a series of control measures [19], [20]. These measures include deploying security personnel at critical locations, erecting temporary barriers for isolation and diversion, implementing one-way organization on key passages, and applying batch release in core areas [21], [22]. For instance, during major holidays, the Shanghai Bund area adopts batch release, passage direction regulation, south-entry north-exit routing, and reduction of head-on conflicts [23]; similarly, Nanluoguxiang main street in Beijing implements one-way passage measures (south entry, north exit) during peak periods [24], consistent in control mechanism with the Shanghai Bund approach.

However, current management practices in urban high-pedestrian-flow areas remain heavily reliant on manual experience and static contingency plans, and still fall short in scientific evaluation and fine-grained optimization [25]. First, many control schemes are derived from historical experience and lack a quantitative characterization of how passage direction settings, entrance release capacities, and phase-dependent behavioral preferences shape crowd dynamics. Second, existing methods tend to rely on single-point statistics or local density monitoring, which makes it difficult to capture the dynamic coupling among "passage direction rules–entrance flow control–local optimal directions–collective density evolution–exit load distribution" [26]. Third, in open scenic areas with multiple entrances, passages, and stages, neglecting visitor route preferences and phase-transition behaviors can lead to misjudgment of passage loads and high-risk locations. Fourth, many simulation- and optimization-based methods treat the crowd model as a black-box evaluator and fail to exploit the structural information embedded in passage geometry, one-way rules, internal entrance capacities, and multi-route subpopulations [27], which renders control-strategy search inefficient [28]. Finally, although evacuation-model verification and validation studies have emphasized component-level testing and emergent-behavior validation [29], existing crowd-control optimization studies still lack an evaluation structure that explicitly separates mechanism diagnostics from strategy-level management objectives.

To address the above issues, this paper proposes a macroscopic crowd modeling and control optimization framework for the "Bund sightseeing platform–multi-stepped passage" open scenic area scenario. The framework connects multi-stage pedestrian movement, passage direction management, geometric guidance, and entrance release control in a unified simulation-and-optimization workflow. It aims to provide a quantitative basis for comparing candidate crowd-control strategies under limited simulation budgets, rather than relying only on empirical rules or post-event assessment.

The main contributions of this paper are as follows.

1) We formulate a multi-stage, multi-route macroscopic model for the open scenic-area class represented by the Bund sightseeing platform and its connected passages. Unlike single-origin–destination or single-potential Hughes-type formulations, the model lets entry, sightseeing, exit, return movement, and route redistribution arise through shared congestion.
2) We develop a Bellman–conservation-law operator that combines hard direction rules, anisotropic mobility guidance, and internal entrance-rate metering. This unified macroscopic operator represents one-way, two-way, and closed channels, channel pre-alignment, open-but-metered entrances, and upstream waiting under mass-consistent dynamics.

3) We establish an evaluation protocol that separates behavioral-mechanism diagnostics from strategy-level management objectives. The protocol first checks whether direction constraints and geometric guidance behave as intended, and then compares policies by efficiency, high-density exposure, load balance, entrance waiting, and control smoothness.
4) We design a hierarchical constrained mixed-variable black-box optimization framework (HCMBO) for joint direction and entrance-rate control. By exploiting the direction–capacity structure and applying unified high-fidelity rechecking, HCMBO achieves the best mean objective among representative mixed-variable baselines under the tested budget.

II. Related Work

A. Massive Crowd Management and Control

Crowd management in urban centers has been extensively studied from control-theoretic and optimization perspectives, aiming to enhance safety and efficiency through boundary regulation, facility layout, and intelligent decision-making [30], [31].

In control-theoretic approaches, Zhong et al. [32] designed boundary control strategies for urban traffic flow networks to ensure convergence to uncongested states under disturbances, with stability analysis based on Lyapunov methods [33]. For one-dimensional crowd dynamics, Wadoo and Kachroo [34] employed advection and diffusion control to prevent blocking and shock waves during evacuation. Addressing multi-directional disturbance propagation, Qin [35] proposed a unit sliding mode controller based on integral barrier Lyapunov functions to ensure boundedness of internal state variables, demonstrating strong disturbance rejection capabilities. For heterogeneous corridor regulation, Zhu et al. [36] combined policy learning with neural networks to learn optimal feedback controllers that adjust inflow rates and free-flow speeds, achieving balanced density distribution and reduced congestion.

Physical infrastructure regulation has also proven effective. Studies have shown that strategically placed obstacles can significantly improve outflow, velocity, and local density in merging areas [25], [37]. Yang et al. [38], [39] employed Model Predictive Control (MPC) to optimize barrier lengths at subway bottlenecks, later extending control variables to entrance limiters, gates, and escalator directions for dynamic, real-time crowd regulation. For evacuation guidance, Carmona and Paricio-Garcia [40] proposed dynamic exit-choice recommendations using multinomial Logit models. However, their approach relies on discrete speed instructions (e.g., "stop," "walk," "run") transmitted via electronic wristbands [41], which poses implementation challenges in open urban scenarios.

Recently, intelligent optimization and deep reinforcement learning have emerged as promising alternatives. Liao et al. [42] formulated fence layout as a subset selection problem, while subsequent work [28] proposed a dynamic crowd inflow control system using LSTM networks to capture temporal features and Proximal Policy Optimization (PPO) [43] for real-time entrance flow regulation. Differential Evolution (DE) [44] has also been applied to optimize guardrail layouts and flow guidance in real-world scenarios such as Chengdu East Railway Station, using average travel time and crowd pressure as multi-objective criteria [45].

B. Macroscopic Crowd Modeling

Macroscopic crowd models treat pedestrian flows as continuous media, offering computational efficiency and analytical tractability compared to microscopic approaches. Henderson [46] first applied fluid dynamics to pedestrian traffic in the early 1970s, comparing crowd motion to the behavior of gas or fluid particles.

Hughes [47] pioneered the first-order continuum theory for pedestrian flow, introducing the classical model in which pedestrian walking speed is governed by local density through a fundamental diagram, while movement direction follows the steepest-descent path of a potential function representing the remaining travel time to exits. This framework elegantly captures the interplay between congestion and route choice. Ling et al. [48] subsequently generalized the Hughes formulation and developed an efficient numerical solution combining a weighted essentially non-oscillatory (WENO) scheme for the conservation law with a fast sweeping method for the eikonal equation.

Several extensions have enriched the descriptive capability of macroscopic models. Nonlocal flux models were proposed by Colombo et al. [49], replacing pointwise density with a neighborhood-averaged quantity in the velocity function to account for anticipatory behavior. This direction was further extended to incorporate anisotropic perception fields and boundary effects from walls and obstacles, with high-resolution numerical schemes and well-posedness results suitable for realistic architectural configurations [50]. To enforce the hard incompressibility constraint—that pedestrian density cannot exceed a physically admissible maximum—Maury et al. [51] introduced the projection model, which formulates the actual velocity as a least-squares projection of the desired velocity onto the set of feasible (non-overlapping) configurations. This framework was later recast as a Wasserstein gradient flow with density constraints [52], using the Jordan–Kinderlehrer–Otto (JKO) scheme [53] to discretize the motion while guaranteeing that density respects an upper bound, making it well-suited for high-density congestion scenarios.

To overcome non-physical behaviors such as supersonic information propagation that may arise in first-order models, Aw and Rascle [54] introduced a second-order model based on a generalized momentum conservation law that preserves anisotropy. The Aw–Rascle (AR) framework was subsequently adapted to pedestrian flows [55], where the congestion pressure term successfully reproduces characteristic phenomena including stop-and-go waves and spontaneous lane formation.

Beyond the classical Hughes-type framework, extensions have incorporated control and guidance mechanisms relevant to crowd management. Anisotropic formulations using mobility tensors have been developed to represent channel geometries and directional preferences [56], [57], allowing for continuous geometric guidance without explicit prohibition. Constrained Hamilton-Jacobi-Bellman (HJB) formulations have been employed to model strict directional constraints such as one-way corridors, with the Bellman optimality principle providing a natural framework for unifying anisotropic geometric guidance with discrete directional constraints.

Despite these advances, existing approaches often treat control parameters as fixed or manually tuned. The joint optimization of discrete channel direction configurations and internal entrance capacity profiles remains underexplored, particularly for large-scale urban pedestrian zones with multiple behavioral stages and heterogeneous route choices.

III. Methodology

A. Problem Definition and Overall Framework

We address a multi-channel crowd management problem in open scenic areas. The proposed method combines a Bellman–conservation-law crowd dynamics model [47], [48] with a joint direction-capacity control formulation. The controllable management variable is defined as

$$z = (s, q) \in \mathcal{Z}, \tag{1}$$

where s denotes the discrete direction states of the channels, and q denotes the time-varying internal entrance capacity profile. The behavioral preference parameter $\hat{p}$ and the geometric guidance parameter η_0 are treated as fixed model inputs. For a given control variable z, the lower-level simulator returns density fields, velocity fields, entrance waiting, and realized channel throughputs. The upper-level optimizer then evaluates efficiency, safety, load balance, waiting, and smoothness metrics, and searches for a better control policy under a limited simulation budget. The overall method framework is shown in Fig. 2.

The key modeling assumption is that direction rules and entrance capacities should be optimized jointly. The direction configuration s changes the local feasible direction set, whereas the entrance capacity q changes the realized admitted flux at channel entrances. These two controls jointly affect channel loading, local high-density exposure, and upstream waiting. Therefore, we formulate crowd management as a mixed-variable simulation-based optimization problem rather than as two independent subproblems, consistent with recent simulation-based passenger-flow and entrance-control studies [28], [58], [59].

B. Bellman–Conservation-Law Crowd Dynamics

1) Conservation Law and Speed Function: We consider a bounded two-dimensional domain $\Omega \subset \mathbb{R}^2$ representing the pedestrian walking area. Let $\Omega_w \subset \Omega$ denote the walkable region, $\rho(x,t)$ the total crowd density, and $\mathbf{v}(x,t)$ the local mean velocity. Mass conservation is governed by

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0. \tag{2}$$

The speed magnitude is determined by local congestion. We adopt the Greenshields-type speed function

$$f(\rho) = v_{\max}\left(1 - \frac{\rho}{\rho_{\max}}\right)_+, \tag{3}$$

where $v_{\max}$ is the free-flow speed, $\rho_{\max}$ is the maximum admissible density, and $(\cdot)_+$ denotes nonnegative truncation. This function captures the feedback that pedestrians move close to free-flow speed in low-density regions and slow down as density approaches congestion.

The classical Hughes model assumes that all movement directions are feasible and isotropic [47]. This assumption is insufficient for crowd management scenarios involving one-way rules, temporary closures, entrance metering, and channel-axis alignment. The following subsections introduce admissible direction sets, anisotropic mobility tensors, and internal entrance capacity control into the Bellman–conservation-law framework.

2) Multi-Stage Multi-Route Density Decomposition: Visitors in open scenic areas usually experience several behavioral stages, such as entry, touring, exit, and return. To represent this behavior chain, the total density is decomposed into stage–route subpopulations:

$$\rho(x,t) = \sum_{\sigma=1}^{S} \sum_{r=1}^{R_\sigma} \rho_{\sigma,r}(x,t), \tag{4}$$

where σ denotes the behavioral stage and r denotes the route type within that stage. All subpopulations share the speed function $f(\rho)$ determined by the total density, but they have different target regions, potential functions, and velocity directions.

For subpopulation (σ, r), the density evolves according to

$$\frac{\partial \rho_{\sigma,r}}{\partial t} + \nabla \cdot (\rho_{\sigma,r} \mathbf{v}_{\sigma,r}) = Q_{\sigma,r}^{\mathrm{in}} - Q_{\sigma,r}^{\mathrm{out}}. \tag{5}$$

If pedestrians in subpopulation (σ, r) complete the current stage in target region $G_{\sigma,r}$ and switch to next-stage route k with probability $\hat{p}_{(\sigma,r)\to k}$, the transition term is

$$Q_{(\sigma,r)\to(\sigma+1,k)}(x,t) = \hat{p}_{(\sigma,r)\to k} \kappa_{\sigma,r} \chi_{G_{\sigma,r}}(x) \rho_{\sigma,r}(x,t), \tag{6}$$

where $\kappa_{\sigma,r}$ is the stage-switching rate, with dimension of inverse time, and $\chi_{G_{\sigma,r}}$ is the indicator function of the completion region. The parameter $\hat{p}$ represents historically inferred or scenario-given route preferences and is fixed in the optimization problem [12], [15], [16].

C. Direction Constraints and Geometric Guidance

1) Direction Configuration Variable: Let the direction state of channel c be

$$s_c \in \{+1, -1, 0, \varnothing\}, \tag{7}$$

where $+1$, -1, 0, and $\varnothing$ denote reference forward one-way, reference reverse one-way, bidirectional, and closed states, respectively. If $\tau_c(x)$ is the reference tangent direction of

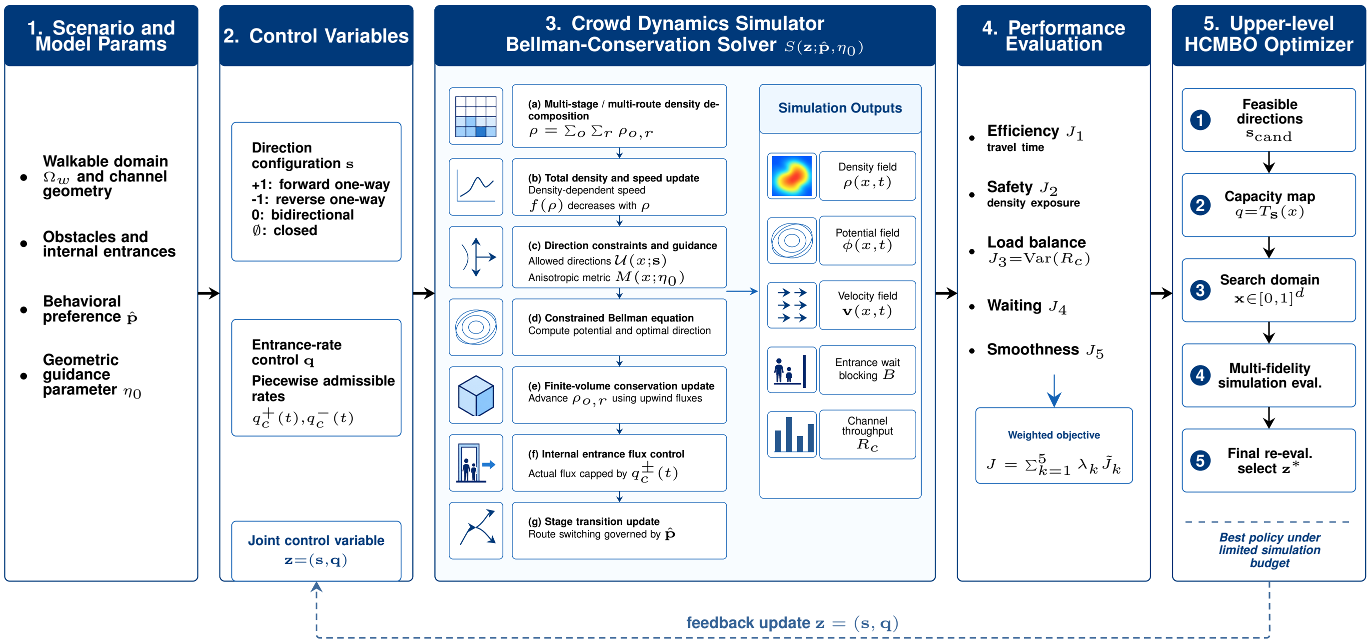

Fig. 2. Method framework for joint direction and entrance-rate optimization. Fixed scenario inputs, behavioral preferences, and geometric guidance parameters define the lower-level Bellman–conservation-law crowd simulator. The controllable direction configuration and entrance-rate profile are jointly evaluated by efficiency, safety, load-balance, waiting, and smoothness objectives, and are updated by the upper-level HCMBO optimizer.

channel c, the local admissible direction set in the channel region is defined as

$$U_c(x;s_c) = \begin{cases} \{\tau_c(x)\}, & s_c = +1, \\ \{-\tau_c(x)\}, & s_c = -1, \\ \{\tau_c(x), -\tau_c(x)\}, & s_c = 0, \\ \varnothing, & s_c = \varnothing, \end{cases} \quad x \in \Omega_c. \tag{8}$$

Thus, s_c changes the local feasible direction set. A closed channel has no admissible direction, a one-way channel retains only one reference tangent, and a bidirectional channel retains both opposite tangential directions.

2) Anisotropic Mobility Tensor: The admissible set $U(x;s)$ represents hard feasibility, but it does not capture the soft geometric guidance induced by narrow channels. Pedestrians approaching a channel often pre-align with the channel axis before entering it. To represent this effect, an anisotropic mobility tensor $M(x;\eta_0)$ is introduced [50], [56], [57]. Equivalently, it is the inverse of a local travel-cost metric. For channel c, let τ_c and n_c denote its tangent and normal directions. The mobility tensor is defined as

$$M_c(x;\eta_0) = \beta_c \left(\eta_0\, \tau_c \tau_c^\top + n_c n_c^\top\right), \qquad \eta_0 \geq 1, \tag{9}$$

where β_c is a channel weight and η_0 controls the mobility preference strength along the channel axis. Larger η_0 increases tangential mobility, equivalently reducing the effective cost of tangential motion and promoting upstream pre-alignment. The parameter η_0 is fixed and is not optimized as a control variable.

3) Constrained Bellman Equation: Given $U(x;s)$ and $M(x;\eta_0)$, the potential function $\phi(x,t)$ represents the minimum remaining cost to the corresponding target region. The discrete Bellman update is

$$\phi(x) = \min_{\mathbf{u}\in U(x;s)} \left[\phi(x + \Delta x\, \mathbf{u}) + \frac{\Delta x}{f(\rho(x,t))} \frac{1}{\sqrt{\mathbf{u}^\top M(x;\eta_0)\mathbf{u}}} \right]. \tag{10}$$

The optimal direction is

$$\mathbf{u}^*(x,t) = \arg\min_{\mathbf{u}\in U(x;s)} (\cdots), \tag{11}$$

and the velocity is

$$\mathbf{v}(x,t) = f(\rho(x,t))\, \mathbf{u}^*(x,t). \tag{12}$$

For the multi-stage multi-route model, each subpopulation (σ, r) uses its own target-specific potential $\phi_{\sigma,r}$ while sharing the same total-density congestion feedback. Thus, direction rules, geometric guidance, and density-dependent congestion are coupled in a single Bellman–conservation-law model.

D. Internal Entrance Capacity Control

Direction rules alone cannot represent the management action in which a channel is open but its entrance must be metered [58], [59], [60]. For each channel, we introduce maximum internal entrance rates

$$q = \{q_c^+(t), q_c^-(t)\}_{c=1}^{C}, \tag{13}$$

where $q_c^+(t)$ and $q_c^-(t)$ denote the maximum admitted rates at the reference forward and reverse entrances of channel c. These controls act on internal channel entrance fluxes rather than on external boundary inflows. In physical units, both the attempted flux $A_c^\pm(t)$ and the capacity bound $q_c^\pm(t)$ are entrance flow rates measured in ped/s.

The direction state and entrance capacity must satisfy the hard consistency constraints

$$\begin{aligned} s_c = +1 &\Rightarrow q_c^-(t) = 0, \\ s_c = -1 &\Rightarrow q_c^+(t) = 0, \\ s_c = \varnothing &\Rightarrow q_c^+(t) = q_c^-(t) = 0, \\ s_c = 0 &\Rightarrow q_c^+(t) + q_c^-(t) \le \bar{q}_c. \end{aligned} \tag{14}$$

Let $A_c^{\pm}(t)$ be the attempted entrance flux induced by free movement. The actually admitted flux is

$$\hat{A}_c^{\pm}(t) = \min\{A_c^{\pm}(t), q_c^{\pm}(t)\}. \tag{15}$$

When $A_c^{\pm}(t) > 0$, the flux across the internal entrance is scaled by

$$\theta_c^{\pm}(t) = \frac{\hat{A}_c^{\pm}(t)}{A_c^{\pm}(t)}. \tag{16}$$

The excess demand $(A_c^{\pm}(t) - \hat{A}_c^{\pm}(t))_+$ remains upstream and is recorded as waiting or queueing mass. This treatment changes channel usage and risk distribution while preserving the mass balance of the conservation-law update.

E. Numerical Solver and Simulation Operator

The walkable region, obstacles, channel areas, and internal entrance interfaces are discretized on a uniform two-dimensional grid. At each time step, the total density is first computed from all stage–route subpopulations and used to update $f(\rho)$. For each subpopulation, the admissible direction set and mobility tensor are then constructed, and the corresponding discrete Bellman equation is solved. After the velocity direction is recovered, an upwind finite-volume update advances the conservation law, while internal entrance fluxes are modified by $\theta_c^{\pm}(t)$ [48]. Finally, stage transition terms are applied to update the subpopulation densities.

For any feasible control variable $z = (s, q)$, the complete numerical workflow defines a lower-level simulation operator

$$(\rho, \phi, \mathbf{v}, B, R) = \mathcal{S}(z; \hat{p}, \eta_0), \tag{17}$$

where B records entrance waiting or blocking and $R = (R_1, \ldots, R_C)$ records realized cumulative throughputs of the channels. The channel balance metric below is computed from the realized throughputs R_c, not from the prescribed capacity limits $q_c^{\pm}(t)$.

F. Evaluation Metrics and Optimization Objective

1) Efficiency, Safety, and Load Balance: Following recent passenger-flow-control and crowd-evacuation simulation-optimization studies [37], [40], [60], the total travel time metric is

$$J_1(z) = \int_0^T \int_{\Omega_w} \rho(x, t)\, dx\, dt. \tag{18}$$

The high-density exposure metric is

$$J_2(z) = \int_0^T \int_{\Omega_w} \mathbf{1}_{\rho(x,t) > \rho_{\text{safe}}}\, dx\, dt. \tag{19}$$

The safety threshold is the density above which exposure is counted, and is fixed at $\rho_{\text{safe}} = 3.5$ ped/m^2 in both the four-channel and Bund scenarios. Let

$$R_c(T) = \int_0^T (\hat{A}_c^+(t) + \hat{A}_c^-(t))\, dt \tag{20}$$

be the realized cumulative throughput of channel c. The channel load imbalance is

$$J_3(z) = \operatorname{Var}(R_1(T), \ldots, R_C(T)). \tag{21}$$

2) Entrance Waiting and Control Smoothness: Entrance waiting is penalized as

$$J_4(z) = \int_0^T \sum_{c=1}^{C} \left(B_c^+(t) + B_c^-(t)\right)\, dt. \tag{22}$$

For piecewise-constant capacity profiles, smoothness is measured by

$$J_5(z) = \sum_{c,k} \left[(q_{c,k+1}^+ - q_{c,k}^+)^2 + (q_{c,k+1}^- - q_{c,k}^-)^2 \right]. \tag{23}$$

Here k indexes the capacity-control time segment. J_4 discourages excessive entrance metering that creates upstream queues, while J_5 discourages abrupt high-frequency changes in capacity profiles [41], [58], [59], [60].

3) Normalization and Scalar Objectives: The raw metrics above have different physical units and numerical scales. Therefore, the reported scalar objectives are computed from dimensionless normalized quantities. Let

$$M_{\text{ref}} = M_0 + M_{\text{in}}, \qquad T_e = T,$$

where M_0 is the initial crowd mass, M_{in} is the cumulative external inflow during $[0, T]$, and T_e is the evaluation horizon. Let $|\Omega_w|$ be the walkable area, $R_{\text{tot}} = \sum_{c=1}^{C} R_c(T)$, and

$$D_R = R_{\text{tot}}^2 \frac{C-1}{C^2}.$$

The basic normalized terms are

$$\bar{J}_1 = \frac{J_1}{M_{\text{ref}} T_e}, \qquad \bar{J}_2 = \frac{J_2}{|\Omega_w| T_e},$$
$$\bar{J}_3 = \frac{J_3}{D_R + \varepsilon_R}, \qquad \bar{J}_4 = \frac{J_4}{M_{\text{ref}} T_e}.$$

For segmented entrance-rate profiles, the smoothness term is normalized by the corresponding entrance-rate bounds:

$$\bar{J}_5 = \frac{1}{N_q} \sum_{c,k,\sigma \in \{+,-\}} \left(\frac{q_{c,k+1}^{\sigma} - q_{c,k}^{\sigma}}{\bar{q}_c^{\sigma} + \varepsilon_q} \right)^2.$$

Here N_q is the number of active finite differences, $\bar{q}_c^{\sigma}$ is the reference upper bound of the corresponding entrance, and ε_R and ε_q only avoid zero denominators. Thus, $\bar{J}_1$ is a relative system-residence measure. The term $\bar{J}_2$ is the average hard exposure or soft safety-risk intensity over the walkable space-time domain [7]. The term $\bar{J}_3$ is zero for perfectly balanced realized channel use and approaches one when all realized channel throughput concentrates on one channel.

The reported terms use scale factors

$$\tilde{J}_k = \frac{\bar{J}_k}{a_k}. \tag{24}$$

TABLE I
RAW UNITS OF THE OBJECTIVE TERMS AND THEIR NORMALIZERS

Term	Raw unit	Normalizer (same unit)	Reported
J_1	ped·s	$M_{\mathrm{ref}}T_e$ (ped·s)	dimensionless
J_2	m^2·s	$\lvert\Omega_w\rvert T_e$ (m^2·s)	dimensionless
J_3	ped^2	D_R (ped^2)	dimensionless
J_4	ped·s	$M_{\mathrm{ref}}T_e$ (ped·s)	dimensionless
J_5	$(\mathrm{ped/s})^2$	$(\bar{q}_c^{\sigma})^2$ $(\mathrm{ped/s})^2$	dimensionless

The main experiments use

$$(a_1, a_2, a_3, a_4, a_5) = (1, 10^{-3}, 1, 1, 1).$$

Therefore, the reported $\tilde{J}_2$ values are scaled normalized safety terms and should not be read directly as space-time percentages; the basic normalized value is $\bar{J}_2 = 10^{-3}\tilde{J}_2$. Given weights $\lambda = (\lambda_1, \ldots, \lambda_5)$, the scalarized objective is

$$J(z) = \sum_{k=1}^{5} \lambda_k \tilde{J}_k(z), \tag{25}$$

with the main-experiment setting

$$(\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5) = (1, 1, 1, 1, 0.1).$$

and the optimization problem is

$$\min_{z \in \mathcal{Z}} J(z), \qquad z = (s, q). \tag{26}$$

The paper solves this fixed-weight scalarized management problem. Different weights can represent different management preferences, but the algorithm itself does not depend on one specific objective term being independently optimal. Although J, $J^{(3)}$, and $\tilde{J}_k$ are dimensionless ranking quantities, their interpretation is anchored by the normalization above [45], [61]. When physical interpretation is needed, the experiments also report raw or ratio-based diagnostics such as channel-flow shares, system mass, and density peaks. Table I lists the physical units of the raw objective terms and their normalizers; in each case the term and its normalizer share the same unit, so every reported $\tilde{J}_k$ is dimensionless.

For mechanism-verification experiments and fixed-strategy diagnostic experiments that only test direction rules, geometric guidance, or fixed entrance-rate responses, we use the reduced three-term objective

$$J^{(3)}(z) = \lambda_1 \tilde{J}_1(z) + \lambda_2 \tilde{J}_2(z) + \lambda_3 \tilde{J}_3(z). \tag{27}$$

In these experiments, the entrance-waiting term $\tilde{J}_4$ is either reported only as a diagnostic quantity or is not applicable, and the smoothness term $\tilde{J}_5$ is meaningful only for segmented capacity-profile optimization. Equivalently, reduced-objective experiments use $\lambda_4 = \lambda_5 = 0$. The numerical values of J and $J^{(3)}$ are not compared across experiment groups.

G. Hierarchical Constrained Mixed-Variable Black-Box Optimization

Since s is discrete, q is continuous or piecewise continuous, and each objective evaluation requires the expensive simulator $\mathcal{S}$, the resulting problem is an expensive mixed-variable black-box optimization problem [62], [63], [64]. We propose HCMBO, a Hierarchical Constrained Mixed-variable Black-box Optimization method, to exploit the structural relation between direction variables and capacity variables.

1) Direction Candidates and Capacity Parameterization: HCMBO first generates a direction candidate set $\mathcal{S}_{\mathrm{cand}}$ satisfying operational and connectivity constraints. For a fixed direction configuration s, the entrance capacity profile is represented by a low-dimensional continuous vector $x \in [0,1]^d$ and mapped to the actual capacity control by

$$q = T_s(x), \qquad x \in [0,1]^d. \tag{28}$$

The mapping T_s automatically applies direction masks, capacity upper bounds, closure constraints, and smoothness restrictions. For example, if a channel is set to reference forward one-way, the reverse entrance capacity is set to zero; if a channel is closed, both directional capacities are set to zero. Thus, the optimizer searches in the continuous x-space while all candidates are mapped to capacity profiles consistent with the discrete direction configuration.

2) Structured Capacity Search: For each direction candidate s, HCMBO performs structured search in the corresponding capacity parameter space. The search uses previously evaluated samples to construct surrogate information for objective values and feasibility, and preferentially selects capacity candidates that satisfy constraints and may improve the incumbent objective. This hierarchical design avoids blind sampling in the full mixed-variable space and separates discrete direction structure from continuous capacity structure.

Multi-fidelity evaluation can be used during search. Low- or medium-fidelity simulations help identify clearly poor capacity profiles, whereas high-fidelity simulations are reserved for final candidate rechecking and ranking. To avoid discarding good long-horizon strategies due to short-horizon bias, low-fidelity results are used only as auxiliary evaluation signals and do not directly determine the final optimum.

3) High-Fidelity Rechecking and Final Selection: All finalist candidates are re-evaluated under a unified high-fidelity setting. Let $\mathcal{C}_{\mathrm{HF}}$ be the candidate set selected for high-fidelity rechecking. The final output is

$$z^* = \arg\min_{z \in \mathcal{C}_{\mathrm{HF}}} J_{\mathrm{HF}}(z). \tag{29}$$

This mechanism prevents surrogate error, low-fidelity bias, or intermediate search ranking from directly becoming the final performance claim.

Algorithm 1 gives the reproducible HCMBO procedure used in the experiments. The loop terminates after the finite optimization budget has been allocated across direction candidates, and the reported optimum is selected only after the high-fidelity rechecking stage.

Compared with generic mixed-variable optimization, HCMBO explicitly exploits direction-capacity consistency. The direction variable determines the effective feasible structure of the capacity variable, and the capacity variable determines the continuous response of entrance release, waiting, and channel load. This structured treatment reduces invalid candidates and concentrates the finite simulation budget on feasible policies with clear management meaning.

IV. NUMERICAL EXPERIMENTS

We validate the proposed model and optimizer on a four-channel scenic-platform scenario abstracted from the Bund

Algorithm 1 HCMBO for Joint Direction and Entrance-Rate Control

Require: Direction constraints Ω_s; simulator $\mathcal{S}$; objective J; gate-rate bounds $\bar{q}$; budget B; initial design size n_0; pool size P; LCB coefficient β; high-fidelity shortlist size K.
Ensure: Best high-fidelity policy z^*.

Generate $\mathcal{S}_{\text{cand}}$ by enumerating $s \in \Omega_s$ and retaining only configurations satisfying open-channel and connectivity constraints.
Rank $\mathcal{S}_{\text{cand}}$ using the prescribed priority templates and direction proxy score, and then truncate it to the direction-candidate limit.
Set $b_s = \lfloor B/|\mathcal{S}_{\text{cand}}| \rfloor$ and assign one extra evaluation to the first $B \bmod |\mathcal{S}_{\text{cand}}|$ directions.
Initialize the global evaluation log $\mathcal{D} \leftarrow \emptyset$.
for all $s \in \mathcal{S}_{\text{cand}}$ **do**
    Construct $T_s : [0,1]^{d_s} \to q$ by applying the direction mask, gate bounds $\bar{q}$, and time segmentation.
    Initialize $\mathcal{D}_s \leftarrow \emptyset$ and generate X_s^0 from high-, medium-, and low-rate profiles plus random fill-up points.
    for all $x \in X_s^0$ **do**
        if $|\mathcal{D}_s| < b_s$ **then**
            Evaluate $z = (s, T_s(x))$ with the optimization-fidelity simulator and append $(x, z, J(z))$ and feasibility metrics to $\mathcal{D}_s$ and $\mathcal{D}$.
        end if
    end for
    while $|\mathcal{D}_s| < b_s$ **do**
        Let x_s^{best} be the evaluated point with the lowest objective in $\mathcal{D}_s$.
        Draw $\mathcal{P}_s$ from P uniform samples in $[0,1]^{d_s}$ and $\max(4, \lfloor P/4 \rfloor)$ Gaussian perturbations of x_s^{best}, clipped to $[0,1]^{d_s}$.
        Set the kernel length scale $\ell_s = \max\{0.18, 1/\sqrt{d_s}\}$.
        for all $u \in \mathcal{P}_s$ **do**
            Estimate $\hat{\mu}_s(u)$ by weighted averaging with $w_i(u) = \exp(-\|x_i - u\|_2^2/(2\ell_s^2))$ over $(x_i, z_i, J_i) \in \mathcal{D}_s$.
            Set $\hat{\sigma}_s(u) = \min\{1, \min_{(x_i,z_i,J_i)\in\mathcal{D}_s} \|u - x_i\|_2/\sqrt{d_s}\}$.
            Compute $a_s(u) = \hat{\mu}_s(u) - \beta\, \text{std}(\mathcal{D}_s)\hat{\sigma}_s(u)$.
        end for
        Select $x_{\text{next}} \leftarrow \arg\min_{u\in\mathcal{P}_s} a_s(u)$.
        Evaluate $z_{\text{next}} = (s, T_s(x_{\text{next}}))$ and append the result to $\mathcal{D}_s$ and $\mathcal{D}$.
    end while
end for
Select the top K unique controls $\mathcal{C}_{\text{HF}}$ from $\mathcal{D}$ by the optimization-fidelity objective.
for all $z \in \mathcal{C}_{\text{HF}}$ **do**
    Re-evaluate z under the unified high-fidelity simulator setting to obtain $J_{\text{HF}}(z)$.
end for
return $z^* = \arg\min_{z\in\mathcal{C}_{\text{HF}}} J_{\text{HF}}(z)$.

sightseeing platform setting. The scenario consists of an inflow region, a Bund sightseeing platform and exit region, a central wall, and four connecting channels: upper, middle, lower-middle, and bottom channels. Throughout this paper, channel denotes a model flow entity indexed by c, whereas passage refers to a real-world walkway at the Bund site that a channel abstracts. The experiments are organized as mechanism validation, directional-response analysis, entrance-rate response, horizontal optimizer comparison, and HCMBO ablation.

A. Experimental Setup

The domain is discretized on a 128×96 grid. To report results in physical units, simulation quantities are mapped through a representative calibration based on three reference scales: a grid cell size of $\Delta x = 1.0$ m, so that the domain spans 128 m $\times$ 96 m; a free-flow walking speed of 1.5 m/s; and a jam density of 5.0 ped/m^2. The induced time step is then $\Delta t = 0.2$ s, and pedestrian counts follow by integrating density over physical area, so that a fully occupied cell holds $\rho_{\max}\Delta x^2 = 5$ ped. As a coherence check, a fully packed domain of this size would hold at most $\rho_{\max} \times 128 \times 96 \approx 6.1 \times 10^4$ ped, which bounds all reported pedestrian counts. These physical units are adopted only for readability and reflect a representative assumption; they do not represent any specific venue.

The channel width is 4 m, and the controlled internal entrance interface width is 16 m. The free-flow speed is $v_{\max} = 1.5$ m/s, the maximum density is $\rho_{\max} = 5.0$ ped/m^2, the initial density is $\rho_{\text{init}} = 2.2$ ped/m^2, and the CFL factor is 0.9. Mechanism and response experiments use the corresponding validation horizons. The optimizer comparison uses high-fidelity rechecks with 1600 simulation steps, a horizon of 320 s (i.e. $\Delta t = 0.2$ s), and Bellman updates every five steps.

Final optimizer rankings are based on the high-fidelity objective in Section III-F. Unless otherwise stated, optimization experiments are ranked by the full five-term scalar objective J after high-fidelity rechecking. Mechanism-validation, directional-response, fixed-rate response, and supplementary transfer diagnostics use the reduced objective $J^{(3)} = \lambda_1\tilde{J}_1 + \lambda_2\tilde{J}_2 + \lambda_3\tilde{J}_3$. For these reduced-objective experiments, $\tilde{J}_4$ is only an entrance-waiting diagnostic or is not applicable, and $\tilde{J}_5$ is not applicable. Equivalently, $\lambda_4 = \lambda_5 = 0$. Therefore, the numerical values of J and $J^{(3)}$ should not be compared across experiment groups. Comparisons are meaningful only within the same objective definition, normalization rule, and weight setting. A high-fidelity best candidate is called feasible if the density-cap removal mass is no more than 2% of the reference mass.

Unless otherwise stated, cumulative flow, waiting mass, unreleased attempted flow, and system mass are reported in pedestrians (ped), obtained by integrating density over the physical cell area. Mass-time quantities are reported in ped·s. Counterflow area-time is reported in m^2·s. Channel-flow shares are computed from realized cumulative channel throughputs, $R_c(T)/\sum_j R_j(T)$. Scalar objectives and normalized terms are dimensionless. Raw cumulative quantities are rounded to three significant digits in the text, while tables retain four decimals for dimensionless objective terms.

B. Mechanism Validation

We first verify whether the two local modeling components have the intended behavioral meanings. The fixed anisotropic mobility tensor $M(x;\eta_0)$ should reshape the channel attraction domain and induce entrance pre-alignment, whereas the admissible direction set $U(x;s)$ should remove prohibited directions and reroute flow through feasible channels. When M is applied

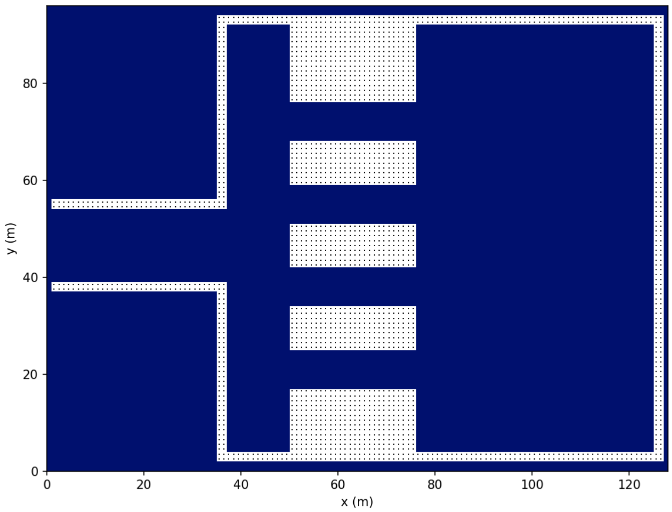

Fig. 3. Four-channel scenic-platform simulation scenario used in the numerical experiments.

to the middle channel, its flow share increases from 48.78% in the baseline to 100.00%. When a one-way rule excludes the originally dominant middle-channel direction, the middle-channel share becomes zero and the flow is redistributed to the upper, lower-middle, and bottom channels with shares 19.25%, 68.76%, and 12.00%, respectively. For the bidirectional validation case, we use two diagnostic quantities. The reverse-direction mass time is

$$\mathrm{RMT} = \sum_n \sum_{i\in\Omega_c} \rho_i^n \mathbf{1}\{\mathbf{u}_i^n \cdot \tau_c < 0\}\Delta x^2 \Delta t,$$

which measures the cumulative reverse-direction mass-time (in ped·s) moving against the prescribed channel direction. The counterflow area-time is

$$\mathrm{CAT} = \sum_n \sum_{i\in\Omega_c} \mathbf{1}\{\rho_{i,+}^n > 0, \rho_{i,-}^n > 0\}\, \Delta x^2 \Delta t,$$

which measures the cumulative area-time (in m^2·s) during which opposite-direction subpopulations coexist in the same channel region. In the baseline bidirectional inflow test, the westbound middle-channel share, RMT, and CAT are 73.18%, 3.33×10^3 ped·s, and 2.43×10^3 m^2·s, respectively. After imposing the eastbound one-way rule, all three quantities are reduced to zero.

Fig. 4 further shows that the combined $U + M$ mechanism is spatially transferable. Changing the guided channel from the upper to the lower channel moves the dominant flow channel and local density hotspot accordingly. These results support using the Bellman–conservation-law simulator as the lower-level evaluator for subsequent strategy optimization.

C. Directional-Control Response

We next examine whether the discrete direction configuration s creates nontrivial system-level trade-offs. The response experiment uses thirteen interpretable direction settings: an all-free baseline, four single-entry cases, four single-return cases, and four one-channel-closure cases. The direction code is ordered as $[T, M, LM, B]$, where E, W, F, and X denote eastbound entry, westbound return, bidirectional free channel, and closure.

Fig. 5 and Table II show that no single direction configuration dominates all objectives. This directional-response experiment summarizes the three reported terms by $J^{(3)}$; since no entrance-capacity curve is optimized, $\tilde{J}_4$ and $\tilde{J}_5$ are not used for ranking. C8 (SR-LM, EEWE) obtains the lowest reduced objective $J^{(3)} = 0.5102$, which is only 0.84% lower than the all-free baseline C1. C1 gives the lowest safety exposure $\tilde{J}_2 = 0.00785$, C3 gives the best efficiency term $\tilde{J}_1 = 0.3141$, and C9 gives the best load-balance term $\tilde{J}_3 = 0.1138$. This non-monotonic response motivates automatic mixed-variable optimization rather than manual selection from a small set of rules.

D. Entrance-Rate Response

Direction rules alone cannot represent the case in which a channel remains open but its internal entrance must be metered. We therefore fix $[T, M, LM, B] = [F, E, W, F]$ and scan four internal entrance-rate levels: no limit, high, medium, and low. The rate upper bound acts on internal entrance crossing flow. When attempted flow exceeds the upper bound, only the admitted flow crosses the interface, and the unreleased attempted flow remains upstream as waiting mass. The entrance binding-time ratio is defined as the fraction of time steps in which attempted entrance flow exceeds the rate upper bound and creates unreleased attempted flow. The unreleased attempted flow is

$$U_{\mathrm{rel}} = \int_0^T \sum_{c=1}^{C} \Big[(A_c^+(t) - \hat{A}_c^+(t))_+ + (A_c^-(t) - \hat{A}_c^-(t))_+\Big]\, dt,$$

in pedestrian-equivalent demand (ped-eq.); because held-back mass re-attempts the interface at later steps, U_{rel} is a cumulative time integral of repeatedly denied demand rather than an instantaneous queue length, and may exceed the number of distinct waiting pedestrians. Its demand ratio is $r_{\mathrm{rel}} = U_{\mathrm{rel}} / \int_0^T \sum_c (A_c^+(t) + A_c^-(t))dt$. This scan is a fixed-rate diagnostic experiment rather than a final optimizer-ranking experiment; the table therefore reports $J^{(3)}$, treats entrance waiting as a diagnostic quantity, and sets the smoothness contribution to zero because each policy uses a fixed rate level.

Table III and Fig. 6 show that tightening the rate upper bound does not simply improve safety. From the no-limit case to the low level, unreleased attempted flow increases from 0 to 1.71×10^3 ped-eq., corresponding to $r_{\mathrm{rel}} = 92.1\%$ of attempted entrance demand. The binding ratio increases from 0 to 0.721, and the scaled normalized safety term $\tilde{J}_2$ increases from 0.174 to 0.466. Thus, q is a genuine continuous control variable that changes system response and must be optimized jointly with s.

E. Horizontal Optimization Comparison

The horizontal comparison evaluates Prior Baseline, Random Search, Pure Simulated Annealing, Enumeration plus Differential Evolution, TPE-Mixed BO, and HCMBO. All

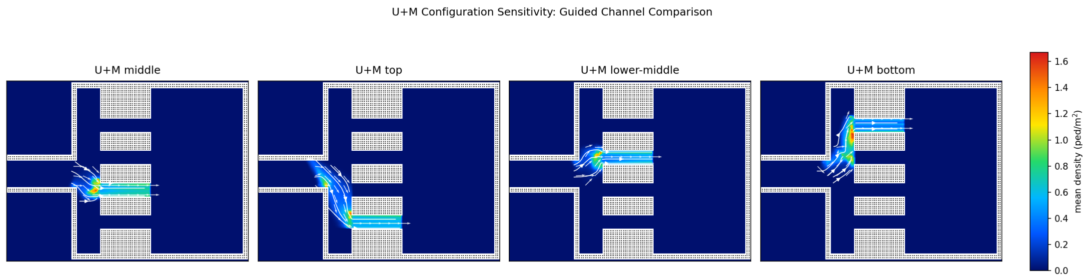

Fig. 4. Spatial transferability of the combined $U+M$ mechanism. Moving the controlled channel induces corresponding migration of dominant flow channels, streamlines, and density hotspots.

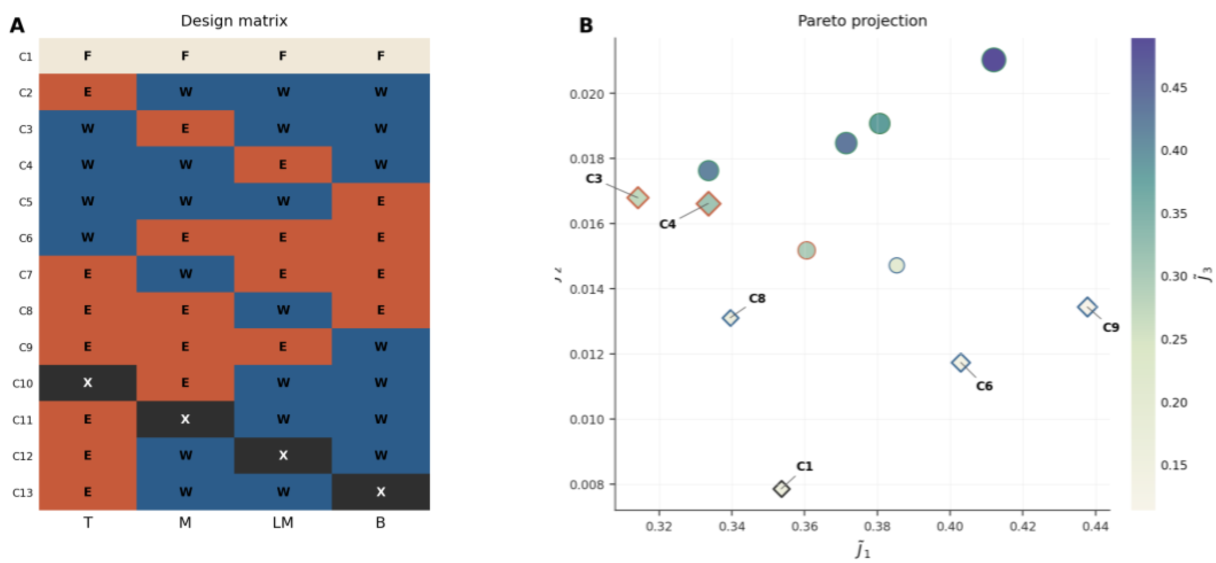

Fig. 5. Directional-control design matrix and objective-space trade-off. Color encodes $\tilde{J}_3$, marker size encodes the reduced dimensionless diagnostic objective $J^{(3)}$, and diamonds indicate non-dominated cases.

TABLE II
NON-DOMINATED DIRECTIONAL-CONTROL CASES

Case	Policy	Code	$\tilde{J}_1$	$\tilde{J}_2$	$\tilde{J}_3$	$J^{(3)}$	Main characteristic
C1	BSL	FFFF	0.3535	**0.00785**	0.1531	0.5145	Lowest exposure
C3	SE-M	WEWW	**0.3141**	0.01680	0.2741	0.6049	Best efficiency but risk-concentrated
C4	SE-LM	WWEW	0.3335	0.01662	0.3131	0.6632	Efficient but load-imbalanced
C6	SR-T	WEEE	0.4029	0.01174	0.1335	0.5482	High return throughput
C8	SR-LM	EEWE	0.3395	0.01311	0.1576	**0.5102**	Lowest reduced objective
C9	SR-B	EEEW	0.4377	0.01344	**0.1138**	0.5649	Most balanced channel loads

All objective terms in this table are dimensionless scaled normalized quantities. The reduced objective uses $\lambda_4 = \lambda_5 = 0$ and is used only for directional-response diagnosis; it is not directly comparable with the full five-term objective in Table IV.

TABLE III
RESPONSE UNDER INTERNAL ENTRANCE-RATE UPPER-BOUND LEVELS

Level	$\tilde{J}_1$	$\tilde{J}_2$	$\tilde{J}_3$	$J^{(3)}$	U_{rel}	r_{rel}	Binding ratio
No limit	0.6274	0.174	0.8838	1.6851	0.00	0	0.000
High	0.6262	0.358	0.8814	1.8652	965.60	70.8%	0.605
Medium	0.6228	0.443	0.8620	1.9276	1317.04	82.6%	0.666
Low	0.6181	0.466	0.8493	1.9337	1711.56	92.1%	0.721

U_{rel} is reported in pedestrian-equivalent demand (ped-eq.), a cumulative time integral of denied entrance demand and not an instantaneous queue length. The safety column is a scaled normalized safety term, not a direct space-time percentage.

optimizers use $B = 400$ candidate evaluations and top-10 high-fidelity rechecks over seeds 11, 23, 37, 51, and 73. The final HCMBO data are taken from the G7-D mainline, whereas the other baselines are taken from the G6 horizontal comparison under the same high-fidelity protocol.

Table IV shows that HCMBO achieves the lowest mean

TABLE IV
HIGH-FIDELITY PERFORMANCE UNDER THE FULL FIVE-TERM DIMENSIONLESS OBJECTIVE

Method	Mean objective	Std.	Median	Best	Worst	Feasible rate
HCMBO	**2.6338**	0.3106	**2.5065**	**2.4495**	3.1851	**100%**
TPE-Mixed BO	2.7403	0.2475	2.6288	2.5180	3.1345	80%
Pure SA	2.9248	0.1386	2.9203	2.7468	3.1049	60%
Random Search	3.0112	0.1319	2.9797	2.8821	3.2287	40%
Enum-DE	3.0597	0.1932	3.1004	2.7329	3.2500	40%
Prior Baseline	5.4774	0.0000	5.4774	5.4774	5.4774	0%

The reported objective is the full dimensionless scalar objective $J = \sum_{k=1}^{5} \lambda_k \tilde{J}_k$ after unified high-fidelity rechecking. These values are not directly comparable with the reduced diagnostic objective $J^{(3)}$ in Tables II, III, and VII.

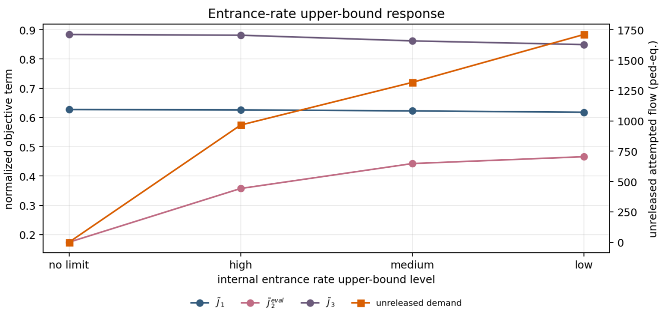

Fig. 6. Objective-term response under internal entrance-rate upper-bound levels. Tightening the upper bound changes safety exposure, load balance, and unreleased attempted flow; the latter is reported in pedestrian-equivalent demand (ped-eq.).

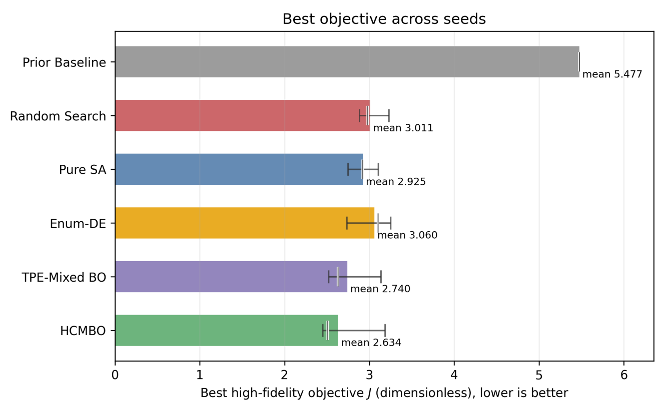

Fig. 7. Best high-fidelity full objective J across seeds. The objective is dimensionless and lower is better. Range bars show the seed-wise spread, and the short vertical mark indicates the median.

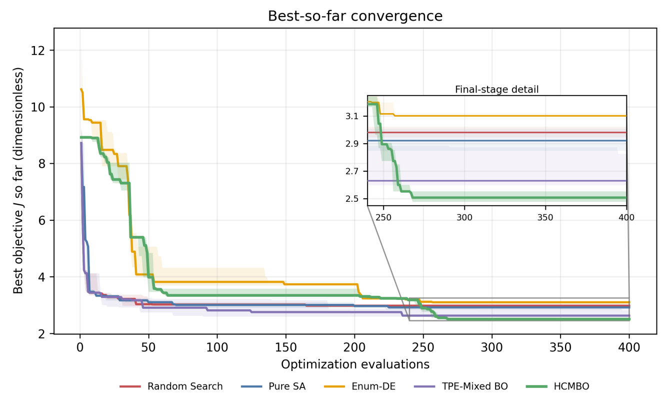

Fig. 8. Best-so-far convergence curves under the same 400-evaluation budget. The inset highlights the final-stage separation among methods.

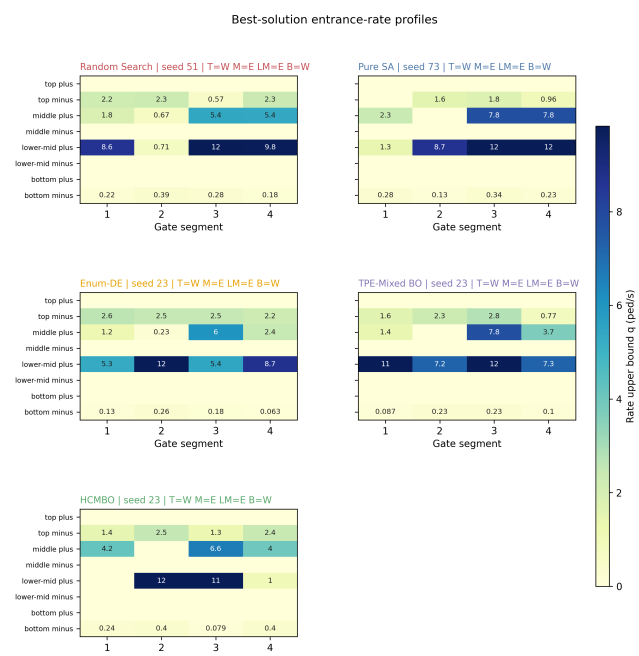

Fig. 9. Best-solution entrance-rate profiles for Random Search, Pure SA, Enum-DE, TPE-Mixed BO, and HCMBO. The common direction structure is $[T, M, LM, B] = [W, E, E, W]$ for several methods, but the segment-wise rate allocations differ substantially.

objective, lowest median objective, lowest best objective, and highest feasible rate. In relative terms, the mean objective of HCMBO is 3.9% lower than that of TPE-Mixed BO, since $(2.7403 - 2.6338)/2.7403 = 3.9\%$. Because this is a normalized scalar objective, the percentage reduction should be interpreted as an improvement in the fixed-weight management score, rather than as a direct reduction in travel time or density exposure alone. Relative to TPE-Mixed BO, HCMBO is better in $4/5$ paired seeds, with an average objective difference of -0.1066. It also outperforms Random Search in all five seeds and outperforms Pure SA and Enum-DE in four of five seeds. The remaining high objective in one HCMBO seed is mainly driven by a larger safety exposure term, indicating a residual need for stronger safety-constraint awareness.

Figs. 7–9 provide complementary visual evidence. TPE-Mixed BO descends quickly in the early evaluations, but

TABLE V
SEARCH-MECHANISM ABLATION UNDER SEED 23

Variant	Best objective	Feasible rate	$\tilde{J}_2$	Unreleased flow (ped-eq.)	Main meaning
HCMBO	**2.4495**	100%	**1.6759**	1.02×10^4	Structured direction-wise rate search
Queue-aware LCB	2.6162	100%	1.8521	$\mathbf{9.64\times10^3}$	Queue-aware acquisition
RF-style constrained BO	2.6162	100%	1.8521	$\mathbf{9.64\times10^3}$	Constraint surrogate variant
Adaptive racing	2.6561	100%	1.9474	1.28×10^4	Stage-wise budget concentration
Adaptive racing + queue-aware	2.6683	100%	1.7862	1.06×10^4	Combined racing and queue signal
Trust-region search	3.0317	100%	1.9224	1.59×10^4	Local trust-region search

Unreleased flow is the cumulative unreleased attempted entrance flow over the full horizon, in pedestrian-equivalent demand (ped-eq.); it integrates repeatedly denied demand and is not an instantaneous queue length.

HCMBO continues to improve in the middle and late stages and reaches the lowest final region. The entrance-rate profiles show that performance differences are not only caused by choosing a discrete direction code; they also arise from detailed segment-wise rate allocation within the same direction structure.

F. HCMBO Ablation

The ablation experiments examine whether the control variables and search mechanisms used by HCMBO are necessary. First, direction and entrance-rate variables must be optimized jointly. Optimizing only direction with a fixed high rate gives best high-fidelity objective 3.3751, while optimizing only rates under prior directions gives 3.3140; both are clearly worse than joint optimization. Second, short-horizon low-fidelity hard screening increases the best objective to 2.9441 and yields an infeasible best candidate, because early simulations do not reliably reflect long-horizon entrance waiting and load redistribution. Third, removing the entrance-waiting penalty slightly reduces the scalar objective to 2.6015, but increases unreleased attempted flow to approximately 1.30×10^4 ped-eq., showing that waiting should remain in the management objective.

Table V shows that the final HCMBO mainline obtains the lowest scalar objective among the tested variants. Queue-aware LCB and RF-style constrained BO reduce unreleased attempted flow to approximately 9.64×10^3 ped-eq., but their scalar objective increases to 2.6162. Thus, queue information is useful for executable and safety-aware extensions, whereas the final mainline should retain structured direction-wise rate search as the primary budget allocation mechanism.

G. Transfer Validation on a Bund-Inspired Scenario

The preceding experiments use a standardized four-channel platform scenario to verify modeling mechanisms and optimization behavior under controlled conditions. As a supplementary validation, we further construct a simplified abstract scenario inspired by the spatial form of the Shanghai Bund waterfront. This experiment does not claim to reproduce all architectural details of the real site. Instead, it preserves the Bund sightseeing platform, the lateral passage connections, the major inflow location, and the dominant movement direction, so as to examine whether an HCMBO-derived policy can still change the load distribution in a more realistic spatial outline.

The computational grid has resolution 512×224. Because this scenario preserves the real Bund outline, its physical scale is calibrated from georeferenced control points that match the scene coordinates to satellite latitude and longitude; a four-point affine fit gives 0.435 m per scene unit with sub-1.5 m residual, which corresponds to a physical cell size of about 1.75 m. The 512×224 grid therefore spans approximately 894 m $\times$ 391 m, the induced time step is $\Delta t \approx 0.44$ s, and pedestrian counts again follow by integrating density over physical area (a cell area of about 3.0 m^2). Pedestrians enter from the outer-side entrance, pass through the middle connecting channels, and move toward the Bund sightseeing platform and its central target region. The inflow-rate multiplier is 3.0, corresponding to an actual inflow rate of about 3.35 ped/s; the maximum density is $\rho_{\max} = 4.0$ ped/m^2, and the stage-switching rate is $\kappa_{\sigma,r} = 32$ in simulation time units. The experimental channels `channel_1`, `channel_2`, `channel_3`, and `channel_4` correspond to the locally named Bund Passages 3, 4, 5, and 6, respectively. Both the controlled and uncontrolled groups use 1600 steps over a physical horizon of about 698 s. For the controlled group, the transferred entrance-rate bounds are multiplied by a capacity-scale factor of 3, providing a less restrictive entrance-capacity setting.

The controlled group uses the best HCMBO policy obtained from the multi-direction high-fidelity recheck. Its reduced high-fidelity objective in the optimization stage is 0.7350, and its direction structure leaves the upper and lower channels bidirectional while assigning opposite dominant directions to the two middle channels. Table VI reports the effective two-segment entrance-rate upper bounds after applying the capacity-scale factor of 3. The rate upper bounds are measured in ped/s.

Table VII compares the controlled and uncontrolled groups under the same high-fidelity protocol. It separates dimensionless normalized terms from raw simulation diagnostics. The reduced objective $J^{(3)}$ decreases from 1.0140 to 0.7410, corresponding to an absolute reduction of approximately 0.2730 and a relative reduction of 26.9%. The main improvement comes from the load-balance term $\tilde{J}_3$, which decreases by 51.9%, from 0.6043 to 0.2908. In contrast, $\tilde{J}_1$ increases by 8.7% and $\tilde{J}_2$ increases by 25.7%. This indicates that the transferred policy changes how pedestrians use the channels, rather than merely reducing all density values uniformly.

The channel-flow shares in Table VIII further explain this reduction. In the uncontrolled group, 82.30% of the channel flow concentrates on `channel_4`, i.e., Bund Passage 6. After applying the HCMBO policy, the flow is redistributed mainly to `channel_2` and `channel_3`, whose combined share

Fig. 10. Real Bund scene and its simplified simulation abstraction. The abstraction retains the Bund sightseeing platform boundary, the main entrance, and the passage connections used for control validation. The satellite basemap was obtained from Google Maps. The right-side local photographs were shot in the Shanghai Bund waterfront passage area by the Tongji Public Safety Experimental Team, June 21, 2026.

TABLE VI
EFFECTIVE ENTRANCE-RATE UPPER BOUNDS IN THE BUND TRANSFER EXPERIMENT

Experimental channel	Local passage name	Entrance side	Segment 1 (ped/s)	Segment 2 (ped/s)
`channel_1`	Bund Passage 3	Forward entrance	3.45	8.38
`channel_1`	Bund Passage 3	Reverse entrance	3.17	0.00
`channel_2`	Bund Passage 4	Forward entrance	8.38	2.36
`channel_2`	Bund Passage 4	Reverse entrance	0.00	0.00
`channel_3`	Bund Passage 5	Forward entrance	0.00	0.00
`channel_3`	Bund Passage 5	Reverse entrance	3.57	6.95
`channel_4`	Bund Passage 6	Forward entrance	5.28	3.03
`channel_4`	Bund Passage 6	Reverse entrance	3.09	2.96

TABLE VII
CONTROLLED AND UNCONTROLLED RESULTS IN THE BUND TRANSFER EXPERIMENT

Dimensionless term	Controlled	Uncontrolled	Relative change
$J^{(3)}$	0.7410	1.0140	-26.9%
$\tilde{J}_1$	0.4145	0.3813	$+8.7\%$
$\tilde{J}_2$	0.0357	0.0284	$+25.7\%$
$\tilde{J}_3$	0.2908	0.6043	-51.9%

Raw diagnostic	Controlled	Uncontrolled	Unit
Final cumulative sink flow	1.33×10^3	1.67×10^3	ped
Final cumulative inflow	2.32×10^3	2.34×10^3	ped
Final system mass	1.53×10^3	1.20×10^3	ped
Peak density maximum	≈ 4.0	≈ 4.0	ped/m^2
Domain-mean density (incl. empty space)	3.47×10^{-3}	3.21×10^{-3}	ped/m^2
Attempted entrance flow	1.49×10^3	–	ped-eq.
Actual entrance flow	1.13×10^3	–	ped
Unreleased attempted flow	364	–	ped-eq.

Sink flow, inflow, system mass, and actual entrance flow are pedestrian counts (ped). Attempted and unreleased entrance flow are cumulative pedestrian-equivalent demand (ped-eq.): they integrate repeatedly denied demand over the horizon and may exceed the number of distinct waiting pedestrians. Entries marked — are not applicable because the uncontrolled case activates no internal metering interface. The domain-mean density is averaged over the full walkable domain, including empty space, and is therefore much smaller than the local peak.

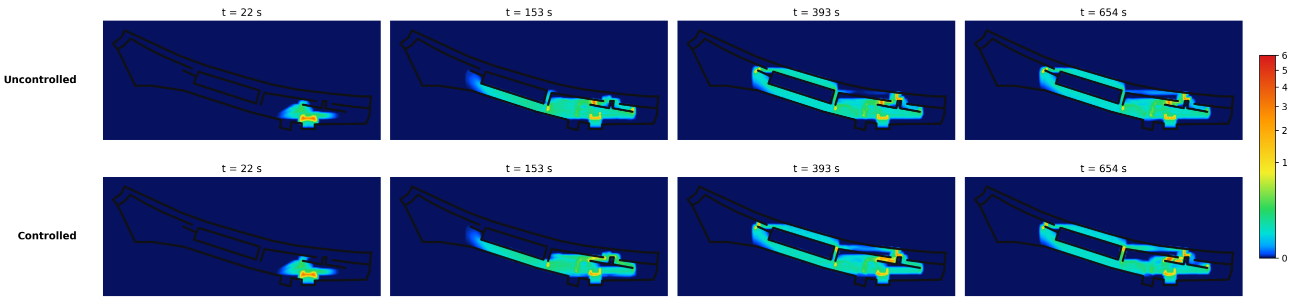

Fig. 11. Density evolution comparison between uncontrolled and controlled cases at approximately 22, 153, 393, and 654 s. The controlled policy redistributes the later-stage flow from the terminal channel toward the middle-channel group.

TABLE VIII
CHANNEL-FLOW SHARES IN THE BUND TRANSFER EXPERIMENT

Channel	Local name	Controlled	Uncontrolled
`channel_1`	Bund Passage 3	2.83%	0.32%
`channel_2`	Bund Passage 4	30.38%	1.35%
`channel_3`	Bund Passage 5	60.96%	16.03%
`channel_4`	Bund Passage 6	5.83%	82.30%

Channel-flow shares are computed as $R_c(T)/\sum_{j=1}^{C} R_j(T)$, where $R_c(T)$ is the realized cumulative admitted throughput of channel c during the simulation horizon.

reaches approximately 91.34%.

Fig. 11 visualizes the density evolution at approximately 22, 153, 393, and 654 s. Both groups initially form a local density concentration near the right-side target region of the Bund sightseeing platform, reflecting the same inflow and target-attraction structure. At later stages, the uncontrolled case remains more concentrated around the terminal channel and the adjacent Bund sightseeing platform area, whereas the controlled case develops a more stable flow band through the middle-channel group. This visual evidence is consistent with the channel-share statistics and confirms that the main effect of the HCMBO policy is load redistribution across channels.

The supplementary transfer experiment also reveals an important trade-off. Compared with the uncontrolled group, the controlled group shows a higher efficiency term ($\tilde{J}_1$ increases from 0.3813 to 0.4145) and a slightly higher safety-exposure term ($\tilde{J}_2$ increases from 0.0284 to 0.0357). At the same time, its cumulative sink flow is smaller (1.33×10^3 versus 1.67×10^3 ped) and its final system mass is larger (1.53×10^3 versus 1.20×10^3 ped). Therefore, the transferred HCMBO policy should not be interpreted as uniformly improving safety and efficiency. Rather, under the present reduced-objective weights, it reduces $J^{(3)}$ mainly by improving channel-load balance, while sacrificing part of the exit throughput and travel efficiency. The capacity-scale factor of 3 reduces unreleased attempted flow to 364 ped-eq., but it does not remove the efficiency cost of the control policy. Overall, the supplementary Bund-inspired experiment shows that the optimized HCMBO policy can still reduce the reduced objective and redistribute flow away from the dominant terminal channel in a simplified real-world spatial abstraction. However, the improvement is accompanied by unreleased attempted entrance flow, lower exit throughput, and higher residual mass in the system. Future deployment-oriented extensions should therefore include explicit throughput constraints, waiting upper bounds, or multi-weight scenario analysis to avoid obtaining policies that improve load balance at the cost of overly strong metering.

V. CONCLUSION

We proposed a macroscopic crowd management framework for open pedestrian zones that jointly models passage-direction rules, geometric guidance, multi-stage route behavior, and internal entrance-rate control. The framework links an interpretable crowd-flow simulator with HCMBO, so that direction settings and entrance-rate profiles can be optimized together under a limited simulation budget.

The experiments show that the model components produce the intended behavioral effects and that the two control dimensions are not interchangeable. Direction settings create non-monotonic trade-offs among efficiency, safety exposure, and load balance, while entrance-rate limits change unreleased attempted flow, binding-time ratios, and local risk patterns. Under five random seeds and unified high-fidelity rechecking, HCMBO obtains the lowest mean full management objective among the tested baselines. It reduces the fixed-weight score by 3.9 percent relative to TPE-Mixed BO and maintains a 100 percent feasible rate. Ablation results indicate that joint direction–rate optimization, avoiding short-horizon hard screening, and using structured direction-wise rate search are central to this performance.

The Bund-inspired transfer experiment further shows that an HCMBO-derived policy can still alter flow organization in a simplified real-world spatial abstraction. In that experiment, the reduced objective decreases by 26.9 percent. This reduction mainly occurs because the policy redistributes flow away from the dominant terminal channel and improves load balance across the middle-channel group. However, this gain is accompanied by higher residual system mass, lower exit throughput, and unreleased attempted entrance flow. Therefore, the transfer result should be interpreted as evidence of mechanistic transferability and load-redistribution capability, not as a ready-to-deploy operational plan for the Bund area.

The current study still has limitations. The experiments are based on simplified geometries, fixed objective weights, and finite random seeds; one HCMBO seed still exhibits relatively high safety exposure; and real crowd calibration, online demand updates, and executable field constraints are not yet included. Future work will extend the framework to richer real geometries, broader demand and seed settings, stronger constraint-aware optimization baselines, multi-weight Pareto analysis, and calibration against field observations.

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